GRP:5 SMP IA-1

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Handwritten Statistics :

1] AIM :

To Calculate mean, median, mode, variance and standard deviation using 25 Temperature observations selected from the Crop Recommendation dataset.

2] Statistical Concepts and Definitions

→ Population: The Complete Collection of Observations under Study. In our project, the population Consists of all 2,200 records in the crop recommendation dataset.

→ Sample: A Subset of observations selected from a population for statistical Analysis. In our project, we use 25 Temperature observations.

→ Representative Sample: A Sample that reflects the important characteristics of the population. Selecting observations from different parts of the dataset helps cover its range.

→ Mean: Mean is the average of all observations in a dataset.

$$\bar{x} = \frac{\sum x}{n}$$

→ Median: Median is the middle Value when observations are arranged in ascending or descending order.

→ Mode: Mode is the Value that occurs most frequently in a dataset.

→ Variance: Variance measures how widely observations are spread around their mean.

→ Standard Deviation: Standard deviation is the positive square root of variance and measures the spread of observations around their mean.

$$SD = \sqrt{Variance}$$

3] Population and Sample:

- Population Size: $N = 2000$
- Sample Size: $n = 25$

For this analysis, 25 Temperatures observations are selected at intervals across the dataset. The temperatures are rounded to one decimal place to simplify hand written Calculations.

- Population Mean Formula:

$$\mu = \frac{\sum X}{N}$$

- Population Variance Formula:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

- Population Standard Deviation:

$$\sigma = \sqrt{\sigma^2}$$

- Sample Mean Formula:

$$\bar{x} = \frac{\sum x}{n}$$

- Sample Variance Formula:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

- Sample Standard Deviation:

$$s = \sqrt{s^2}$$

4] Sample Data:

Observation No.	Temperature [x]
1	20.9
2	23.8
3	25.1
4	17.1
5	23.4
6	31.5
7	32.0
8	29.8
9	31.8
10	29.9
11	22.4
12	29.6
13	29.5
14	33.9
15	16.3
16	26.9
17	27.7
18	22.5
19	11.5
20	28.3
21	26.9
22	25.0
23	24.4
24	23.1
25	23.6
Total	636.9

$n = 25, \quad \sum x = 636.9$

5] Calculation of Mean:

Formula:

$$\bar{x} = \frac{\sum x}{n}$$

Substituting the values:

$$\bar{x} = \frac{636.9}{25}$$

$$\boxed{\bar{x} = 25.476 \approx 25.48^{\circ}C}$$

Therefore, the average Temperature is approximately 25.48°C.

6] Calculation of Median:

Arrange the observations in ascending order:

11.5, 16.3, 17.1, 20.9, 22.4, 22.5, 23.1, 23.4, 23.6, 23.8, 24.4, 25.0, 25.1, 26.9, 26.9, 27.7, 28.3, 29.5, 29.6, 29.8, 29.9, 31.5, 31.8, 32.0, 33.9,

Since the No. of observations is odd:

$$\text{Median Position} = \frac{n+1}{2}$$

$$= \frac{25+1}{2}$$

$$= 13$$

The 13th observation is 25.1.

$$\boxed{\text{Median} = 25.1^{\circ}C}$$

7] Calculation of Mode :

Mode is the Value that Occurs most frequently in a dataset.

In our sample, 26.9 occurs twice, while others occured once.

$$\text{Mode} = 26.9^{\circ}C$$

8] Calculation of Variance:

The mean is : $\bar{x} = 25.476$

The deviations and squared deviations are rounded to two decimal places in this table.

x	$x - \bar{x}$	$(x - \bar{x})^2$
20.9	-4.58	20.94
23.8	-1.68	2.81
25.1	-0.38	0.14
17.1	-8.38	70.16
23.4	-2.08	4.31
31.5	6.02	36.29
32.0	6.52	42.56
29.8	4.32	18.70
31.8	6.32	39.99
29.9	4.42	19.57
22.4	-3.08	9.46
29.6	4.12	17.01

A]

x	$x - \bar{x}$	$(x - \bar{x})^2$
29.5	4.02	16.19
33.9	8.42	70.96
16.3	-9.18	84.20
26.9	1.42	2.03
27.7	2.22	4.95
22.5	-2.98	8.86
11.5	-13.98	195.33
28.3	2.82	7.97
26.9	1.42	2.03
25.0	-0.48	0.23
24.4	-1.08	1.16
23.1	-2.38	5.65
23.6	-1.88	3.52
Total:		685.01 approx

A] Population Variance of the Sample:

Formula:

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$$

Calculation:

$$\sigma^2 = \frac{685.01}{25}$$

$$\sigma^2 \approx 27.40$$

B] Sample Variance:

Formula:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

Calculation:

$$s^2 = \frac{685.01}{25-1}$$

$$s^2 = \frac{685.01}{24}$$

$$\boxed{s^2 \approx 28.54}$$

9] Calculation of Standard Deviation:

A] Population Standard Deviation of the Sample:

Formula:

$$\sigma = \sqrt{\sigma^2}$$

Calculation:

$$\sigma = \sqrt{27.40}$$

$$\boxed{\sigma \approx 5.23^\circ C}$$

B] Sample Standard Deviation:

Formula:

$$s = \sqrt{s^2}$$

Calculation:

$$s = \sqrt{28.54}$$

$$\boxed{s \approx 5.34^\circ C}$$

10] Interpretation:

The mean Temperature of the Sample is approximately 25.48°C, and the Median is 25.1°C. The Mode is 26.9°C. The variance calculated by dividing by n is approximately 27.40, whereas the Sample Variance calculated by dividing by n-1 is approximately 28.54. Their corresponding standard deviations are approximately 5.23°C. and 5.34°C.

These results help us understand the average temperature and the variation in agricultural data. Temperature analysis can help us understand environmental conditions relevant to Crop recommendation.

Important Clarification:

The Values 27.40 and 5.23 are the population variance and standard deviation when treating these 25 observations as the entire group. They are not the variance and standard deviation of all 3,200 records. The Sample variance and standard deviation are the more appropriate estimates when using 25 observations to represent the larger datasets.

11] Conclusion:

Mean, Median and Mode describe the central values of a dataset, while variance and standard deviation measure the spread of the observations. These statistical concepts help analyze agricultural data and provides a foundation for understanding crop recommendation systems.